

Luther Bell

Software Engineer

luthbell4@gmail.com | [LinkedIn](#) | [Portfolio Website](#)

Software Engineer with 4+ years of experience designing and building scalable backend services, REST APIs, and enterprise applications using Java, Spring Boot, .NET, SQL, Kubernetes, and cloud-native technologies. Experienced delivering secure production systems, modernizing legacy applications, and collaborating across engineering teams to ship reliable software.

Experience

Software Engineer

General Motors — March 2022 – Present

- Architected and implemented secure Java Spring Boot REST APIs that replaced manual UI-based CSV exports with scalable programmatic data access for an external engineering team by modernizing a legacy .NET Framework application, implementing OAuth 2.0 authentication, and deploying services to Kubernetes.
- Investigated and resolved recurring production UI and database timeout issues by identifying a missing SQL database index, improving query performance, and enhancing application observability with OpenTelemetry and Datadog, eliminating recurring production incidents related to database timeouts.
- Modernized application infrastructure by rebuilding and migrating an application to Windows Server 2022, automating IIS provisioning and deployments with PowerShell, GitHub Actions, and YAML, transforming a manual deployment process into a one-click deployment workflow.
- Designed and implemented reusable GitHub Actions and Azure DevOps CI/CD pipelines from scratch, automating builds, testing, Docker image creation, artifact packaging, and deployments across 4 enterprise applications while standardizing release workflows and reducing manual deployment effort.
- Extended an existing .NET Framework file ingestion service to support a new external application by implementing parsing logic for a new file format, triggering database updates through stored procedures, and establishing a reusable integration pattern for future partner onboarding.
- Redesigned a .NET Framework integration to migrate from mutual TLS to the OAuth 2.0 Client Credentials flow after an external partner deprecated certificate-based authentication, preventing service disruption while maintaining secure production communication between systems.
- Upgraded enterprise applications from .NET 8 to .NET 10, improving maintainability, strengthening application security, and ensuring long-term platform support.

Education

St. John's University

Bachelor of Science: Computer Science - Cyber Security

AWS Fundamentals Specialization – Coursera (Completed Sept 2025)

Technical Skills

- **Languages:** Java, C#, SQL, Python, TypeScript
- **Backend:** Spring Boot, .NET Framework, REST APIs, FastAPI, Flask
- **Frontend:** TypeScript, React
- **Cloud & DevOps:** Kubernetes, OpenShift, Docker, GitHub Actions, Azure DevOps, AWS

- **Databases:** PostgreSQL, SQL Server, Oracle
- **Testing & Observability:** JMeter, Selenium, TestNG, OpenTelemetry, Datadog, Playwright
- **Tools:** Git, IntelliJ IDEA, VS Code, GitHub Copilot

Personal Projects

AI Resume Screening System — *Python, FastAPI, Hugging Face Transformers, PostgreSQL, REST APIs*

- Built an AI-powered resume screening platform using transformer embeddings to perform semantic similarity matching between resumes and job descriptions.
- Developed a FastAPI backend exposing REST APIs for candidate scoring, explainable match results, and scalable inference workflows.

Food Diary Application — *Java, Spring Boot, PostgreSQL, React, Tailwind CSS*

- Built a full-stack meal tracking application using Spring Boot, PostgreSQL, React, and Tailwind CSS.
- Designed REST APIs, implemented relational data models, and integrated backend services with a responsive React frontend.

AI Chatbot Web App — *Next.js, Flask, Hugging Face Transformers, Tailwind CSS*

- Built a full-stack chatbot using Next.js, Flask, and open-source language models.
- Developed backend APIs to integrate AI models, manage conversation state, and support real-time user interactions.